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Analyzing Patient Characteristics and Lung Cancer Outcomes Pre and Post the 2021 USPSTF Lung Cancer Screening Guidelines

Experience From a Large Academic Institution

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Purpose: In 2021, the US Preventive Services Task Force (USPSTF) revised the guidelines for lung cancer screening (LCS). Numerous studies have examined the effects of the guideline changes on LCS eligibility. Yet, few have focused on their impact on participation and lung cancer outcomes within a clinical LCS cohort.

Materials and Methods: This retrospective cohort study included individuals who underwent low-dose computed tomography (LDCT) screening between July 31, 2013, and August 25, 2023, at a large tertiary academic medical center. Patients were categorized into the preguideline and postguideline revision groups based on LCS eligibility and insurance status. Patient characteristics at baseline screening and lung cancer outcomes were compared.

Results: Among 3929 patients, after implementing the new guidelines, the average monthly volume of new LCS patients nearly tripled (68 vs. 23, $P < 0.05$). Notable reductions were observed in age (mean: 63 vs. 65, $P < 0.05$), pack-years (mean: 35 vs. 44, $P < 0.05$), and PLCOm2012 risk score (mean: 2.9% vs. 4.1%, $P < 0.05$). The proportion of negative LDCT results increased (89.4% vs. 84.5%, $P < 0.05$). The proportion of Black patients decreased (5.1% vs. 6.9%, $P < 0.05$), while Hispanic patients increased from 5.5% to 7.6% ($P < 0.05$). The lung cancer detection rate fell in the postguideline group (0.9% vs. 4.5%, $P < 0.05$), with 8 lung cancers detected among 943 newly eligible patients.

Conclusions: Disparities in LCS participation among racial and ethnic groups were observed following the 2021 USPSTF LCS guidelines, highlighting the need for active outreach and patient education, as guideline revisions alone increase eligibility but do not ensure participation.

Key Words: Lung cancer screening, United States Preventive Services Task Force (USPSTF) guidelines, racial and ethnic disparity

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Lung cancer is the leading cause of cancer-related mortality in both sexes in the United States.[1] Screening with low-dose computed tomography (LDCT) reduces mortality.[2,3] In December 2013, the US Preventive Services

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Task Force recommended annual LDCT screening for high-risk individuals aged 55 to 80 with at least 30 pack-years of smoking intensity who are current smokers or have quit within the past 15 years (grade "B").[4] The Centers for Medicare and Medicaid Services (CMS) provided coverage for LDCT screens beginning February 2015.[5] Due to the USPSTF's grade B recommendation for lung cancer screening (LCS), private insurers are mandated to cover LDCT screening costs.[6] Despite insurance coverage, participation in lung cancer screening remains strikingly low, with only 5.8% of eligible smokers in the United States screened in 2021.[7]

In March 2021, the USPSTF revised its recommendation for LCS based on recent research.[8] The new guidelines reduced the lower limit of the screening age from 55 to 50 years and the minimum smoking history from 30 to 20 pack-years.[9] Accordingly, the CMS expanded the eligibility criteria for Medicare beneficiaries receiving screening LDCT in February 2022. Under the new guidelines, the eligibility for LCS was extended to 14.5 million Americans, representing an increase of 6.5 million individuals compared with the previous guidelines.[10,11] Previous studies suggest that the new guidelines will increase the number of women and racial and ethnic minority groups who are eligible for lung cancer screening.[10,12,13] However, sex, racial, and ethnic disparities may persist without tailored eligibility criteria.[10,13–16]

These studies, whether relying on self-reported national surveys or non-LCS patient cohorts, evaluated the expansion of eligibility for LCS. Still, they did not specifically examine whether these newly eligible patients eventually underwent LDCT screening. Research comparing the changes in patient characteristics and lung cancer outcomes before and after the new LCS guidelines in a clinical LCS program remains scarce.[17]

Following the implementation of the new LCS guidelines, there has been an increase in the number of new patients enrolling in our LCS program. In this study, we assessed whether the near tripling of eligible individuals predicted with new LCS eligibility guidelines would lead to a proportional increase in patient participation in our program. Furthermore, we tested the hypotheses that patients undergoing LCS after the implementation of the updated guidelines would be younger, more racially and ethnically diverse, and have a reduced smoking history, lower PLCOm2012 risk scores, and lower baseline Lung CT Screening Reporting & Data System (Lung-RADS) scores as a result of the lower PLCOm2012 risk scores. Lung cancer-related outcomes (eg, detection rate, stage, and histology) were compared before and after implementing the new LCS guidelines.

MATERIALS AND METHODS

Patient Enrollment

Institutional Review Board approval was obtained at the University of California, Los Angeles to conduct this retrospective cohort study. Informed consent was waived because the risk to patients was minimal. We included patients who underwent their baseline LDCT screening examination at UCLA between July 31, 2013, and August 25, 2023, across 14 geographically distributed sites in our health system within a 30-mile radius of our main campus where LCS is offered. Patients were excluded if they did not meet the appropriate USPSTF LCS guidelines. Patients were referred by their primary care physician or other clinicians for LCS. After shared decision-making, the patient must call to schedule the low-dose CT exam. There were no major changes in patient enrollment during the study period. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cohort studies.

Data Collection

Patient information at the baseline screen was extracted from the electronic health record (EHR) system, including age, primary insurance, smoking status (current or former), smoking intensity (ie, the average number of cigarettes per day), smoking duration, years since quitting (former smokers), and Lung-RADS score. Pack-years were derived by multiplying the average number of packs per day (1 pack = 20 cigarettes) and years smoked. Lung-RADS score was manually assigned to LDCT examinations performed before the release date of Lung-RADS 1.0 by a board-certified radiologist. Smoking data were manually extracted from clinical notes by one of the authors, who has 7 years of experience with data abstraction from EHR. During smoking history data collection, information from the shared decision-making visit was prioritized, followed by the primary care physician visit and other notes. When smoking data were missing from the EHR, we used patients' self-reported smoking history collected from a self-reported questionnaire administered before the LDCT screening examination (stored as a discrete series of the screening examination in our picture archiving and communication system). The questionnaire also collected other variables required by the PLCOm2012 risk model. Race and ethnicity data were included to assess the racial and ethnic proportions of UCLA patients undergoing screening because variations in lung cancer risk have been identified across different racial and ethnic groups. Body mass index (BMI) was calculated using weight and height. A manual chart review addressed missing race, ethnicity, and BMI information from the questionnaire. The 6-year PLCOm2012 risk score was calculated using age, race/ethnicity, education, BMI, history of COPD, personal history of cancer, family history of lung cancer, smoking status, smoking intensity, smoking duration, and years since quitting.[18] The 2013 and 2021 USPSTF guidelines determined LCS eligibility.[4,9] Lung cancer diagnosis was obtained from EHR through manual chart review, and the last date of patient follow-up was August 25, 2023. Prequalified patients were defined as meeting the 2013 USPSTF LCS guidelines. Newly eligible patients were those who met the 2021 but not the 2013 USPSTF LCS guidelines: current smokers or former smokers 15 years since quitting, (1) aged 50 to 55 and 20 to 30 pack-years, (2) aged 50 to 55 and $\geq$30 pack-years, and (3) aged $\geq$55 and 20 to 30 pack-years.

Types of Lung Cancers

A screening result was considered negative if the Lung-RADS score was 1 or 2 and positive if the score was 3 or 4. We defined 6 types of cancers in LCS according to the literature[19,20] and opinions from 2 board-certified thoracic radiologists.

1. Prevalence cancer: The baseline screen is positive, leading to downstream testing and subsequent diagnosis of lung cancer, including new and preexisting cancers if they were observed on a prior nonscreening CT.

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2. Missed prevalence cancer: The lesion is present on the baseline LDCT screen but not described as positive and, therefore, not acted upon.
3. Incidence cancer: The annual follow-up (incidence) screen is positive, prompting downstream testing and lung cancer diagnosis. The cancer occurs following a positive incidence screen but does not include cancers "visible in retrospect."
4. Missed incidence cancer: The lesion is present on the incidence (ie, annual follow-up) LDCT screen but not described as positive and not acted upon.
5. Interval cancer: The screen is negative, but lung cancer is diagnosed before a subsequent screen. Such lesions are not considered screen-detected cancers but cancers that develop between screening intervals.
6. No-screen–related cancer: A lung cancer diagnosis is made more than 18 months from the last screening exam. This designation applies to lung cancers detected after missing one or more recommended annual screens.

Patient Subgroups

Patients were categorized into preguideline and postguideline revision groups based on baseline LDCT screen date, insurance status, and LCS eligibility (Fig. 1). The expanded CMS coverage for LDCT screens was issued almost 1 year after the 2021 USPSTF LCS guidelines (March 9, 2021, for USPSTF and February 10, 2022, for CMS); this affected the insurance coverage for newly eligible LCS patients (ie, current smokers or former smokers 15 y since quitting aged 50 to 55 and 20 to 30 pack-years, aged 50 to 55 and $\geq$30 pack-years, and aged $\geq$55 and 20 to 30 pack-years) who had Medicare and/or Medicaid insurance. Thus, newly eligible Medicare and/or Medicaid-insured patients were in the preguideline revision group if their baseline LDCT screen was performed before February 10, 2022; otherwise, they were in the postguideline revision group. All other patients were in the preguideline revision group if their baseline LDCT screen was completed before March 9, 2021; otherwise, they were categorized into the postguideline revision group. We also considered 2 other approaches for categorizing patients into 2 groups: (1) including an equal number of patients from each period, and (2) including an equal length of time from each period. However, after consulting with a biostatistician, we determined that both approaches would introduce bias. Therefore, we decided to include all patients from both periods.

Statistical Analysis

We plotted the number of newly enrolled LCS patients by month and year to illustrate patient volume variations over the first ten years of implementing LCS at our institution. The average monthly number of new LCS patients was calculated for the 2 groups. Data were summarized using the mean and SD. Subgroup differences in the means were assessed using Welch 2-independent sample t test, Wilcoxon rank-sum test, or the Kruskal-Wallis test. The χ^2 and Fisher exact tests were used for subgroup comparisons of categorical variables. Data analysis was performed using R, version 3.6.1, and Python 3.7.3 (Python Software Foundation). Two-sided $P < 0.05$ was considered significant.

RESULTS

Patient Characteristics

There were 3929 eligible patients in total. Mean (SD) age was 64 (7) years; 1505 patients (38.3%) were female; 2424 (61.7%) male; 18 (0.5%) American Indian or Alaska Native; 318 (8.1%) Asian; 237 (6.0%) Black; 258 (6.6%)

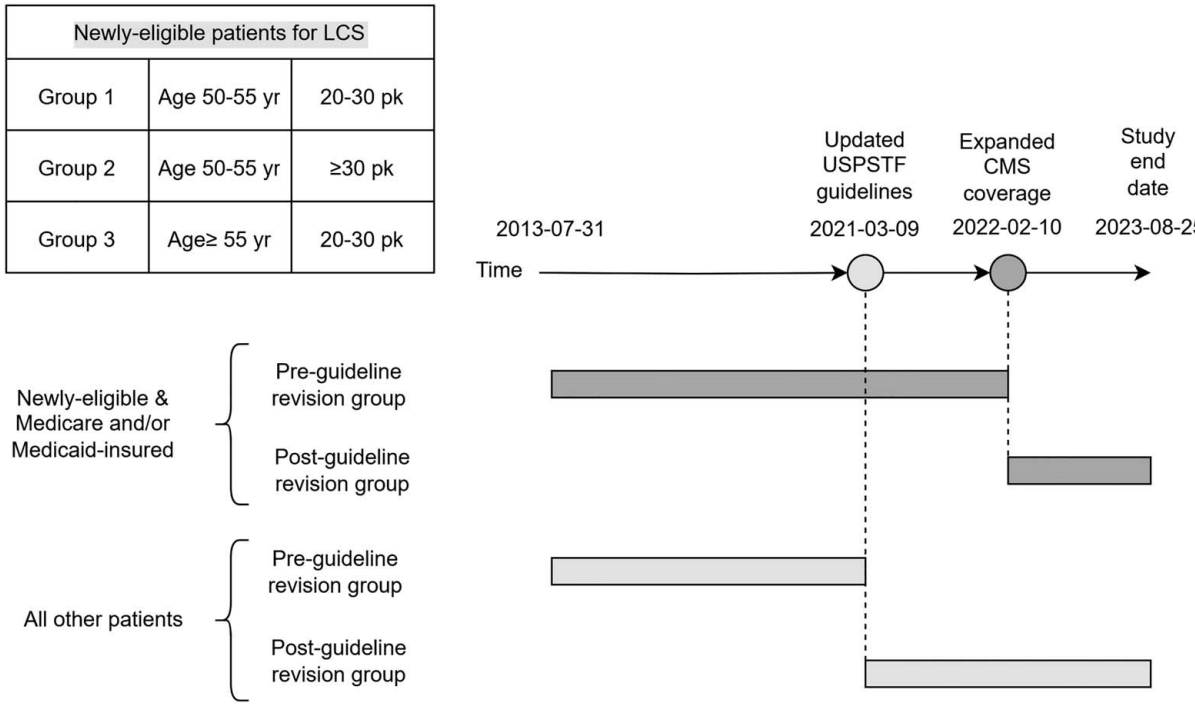


FIGURE 1. Categorization of LCS patients into preguideline and postguideline revision subgroups.

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TABLE 1. Patient Characteristics at Baseline Screen (N = 3929)

Variable	Category	Value
Age [mean (SD)]	—	64 (7)
Sex, n (%)	Female	1505 (38.3)
	Male	2424 (61.7)
Race/ethnicity, n (%)*	American Indian or Alaska Native	18 (0.5)
	Asian	318 (8.1)
	Black	237 (6.0)
	Hispanic	258 (6.6)
	Native Hawaiian or Pacific Islander	1 (<0.1)
	White	2920 (74.3)
	Other	35 (0.9)
	Unknown	142 (3.6)
Education level, n (%)	Less than high school graduate	128 (3.3)
	High school graduate or GED	633 (16.1)
	Some training after high school	137 (3.5)
	Some college or technical school	1049 (26.7)
	College graduate	962 (24.5)
	Postgraduate	666 (17.0)
	Unknown	354 (9.0)
Primary insurance, n (%)†	Medicare and/or Medicaid	1372 (34.9)
	Private or commercial	2497 (63.6)
	Other	56 (1.4)
	Unknown	4 (0.1)
Smoking status, n (%)	Current	1698 (43.2)
	Former	2226 (56.7)
	Unknown	5 (0.1)
Family history of lung cancer, n (%)	Yes	876 (22.3)
	No	3053 (77.7)
Lung-RADS category, n (%)	0	13 (0.3)
	1	779 (19.8)
	2	2637 (67.1)
	3	240 ((6.1)
	4A	153 (3.9)
	4B	78 (2.0)
	4X	29 (0.7)

*Subcategories in other race: more than one race, or other racial and ethnic groups not otherwise stated.

†Subcategories in other insurance: Veterans Administration, self-pay, or other insurance not specified.

Lung-RADS indicates Lung CT Screening Reporting and Data System.

Hispanic; 1 (<0.1%) Native Hawaiian or Pacific Islander; 2920 (74.3%) white; and 35 (0.9%) other race and ethnicity. Most patients [2497 (63.6%)] had private or commercial insurance. A total of 2226 (56.7%) patients were former smokers. Most patients [3416 (86.9%)] had a negative baseline screening result. Additional patient characteristics at the baseline screen are summarized in Table 1.

Patient Volume Over a Decade

Figure 2 delineates the patterns in LCS patient volume by month and year over a decade. The adoption of LDCT screens has increased over time. Screening volume was minimal in 2014 due to the recent initiation of annual LDCT screening recommendations by the USPSTF for eligible individuals. From 2015 to 2020, there were ~20 to 40 new patients monthly. LDCT screening was temporarily paused at our institution during the peak of the COVID-19 pandemic (March to May 2020),[21] leading to a substantial decline in patient volume during these months. Patient volume returned to baseline after the COVID-19 pause. Immediately after the release of the USPSTF LCS guidelines in March 2021, there was a substantial increase in the monthly count of new LCS patients, with the number of new patients per month ranging from ~50 to 100. The average monthly number of new LCS patients almost tripped in the postguideline revision group compared with the preguideline revision group (68 vs. 23, $P < 0.05$).

Patient Characteristics Between Subgroups

The preguideline and postguideline revision groups included 1963 and 1966 patients, respectively. The median follow-up times for the 2 groups were 5.3 (interquartile range: 4.1 to 6.9) and 1.1 (interquartile range: 0.5 to 1.7) years after the baseline screen. Table 2 compares patient characteristics pre and post the new USPSTF guidelines at baseline screening. The mean age was 2 years lower in the postguideline revision group compared with the preguideline revision group (63 vs. 65, $P < 0.05$). There was no difference in sex between the 2 groups, with most patients being male ($P > 0.05$). The proportions of race and ethnicity categories were statistically different between the 2 groups ($P < 0.05$). As opposed to the preguideline revision group, the postguideline revision group had a slightly higher proportion of Hispanic patients (7.6% vs. 5.5%), fewer Black (5.1% vs. 6.9%), and fewer White patients (71.6% vs. 77.0%). Moreover, the postguideline revision group exhibited a higher proportion of patients with less than a college education (51.2% vs. 46.9%, $P < 0.05$) and a higher percentage of patients with private or commercial insurance (72.5% vs. 54.6%, $P < 0.05$). In addition, the postguideline revision group had a lower smoking intensity (19 vs. 22, $P < 0.05$), a lower mean duration of smoking in years (37 vs. 41, $P < 0.05$), lower mean pack-years (35 vs. 44, $P < 0.05$), and a lower mean PLCOm2012 risk score (2.9% vs. 4.1%, $P < 0.05$). A notable increase in the proportion of negative (ie, Lung-RADS 1 or 2) LDCT screening results was observed in the postguideline revision group compared with the preguideline revision group (89.4% vs. 84.5%, $P < 0.05$). Approximately half (48%) of the postguideline revision group patients were newly eligible.

PLCOm2012 Score in Subgroups

In both groups, there were significant differences in the mean PLCOm2012 score for all variables ($P < 0.05$), except for sex in the preguideline revision group (Table 3). The mean PLCOm2012 score was higher in patients aged 65 years and older (pre: 5.6 vs. 2.2, post: 4.8 vs. 1.5, $P < 0.05$), females (post: 3.1 vs. 2.7, $P < 0.05$), Black race (pre: 5.7 vs. 1.9 to 4.3, post: 4.6 vs. 1.1 to 3.1, $P < 0.05$), those with lower education levels (pre: less than high school 6.0 vs. 3.5-5.2, post: some training after high school 7.2 vs. 2.3 to 4.2, $P < 0.05$), those with Medicare and/or Medicaid insurance (pre: 5.1 vs. 3.3 to 3.6, post: 4.6 vs. 0.9 to 2.3, $P < 0.05$), current smokers (pre: 5.0 vs. 3.6, post: 3.7 vs. 2.3, $P < 0.05$), those with a family history of lung cancer (pre: 5.9 vs. 3.6, post: 4.1 vs. 2.5, $P < 0.05$), those with a higher Lung-RADS score (pre: 4X 7.2 vs. 3.4 to 6.6, post: 4A 4.8 vs. 2.1 to 4.6, $P < 0.05$), and prequalified patients (post: 4.2 vs. 1.5, $P < 0.05$).

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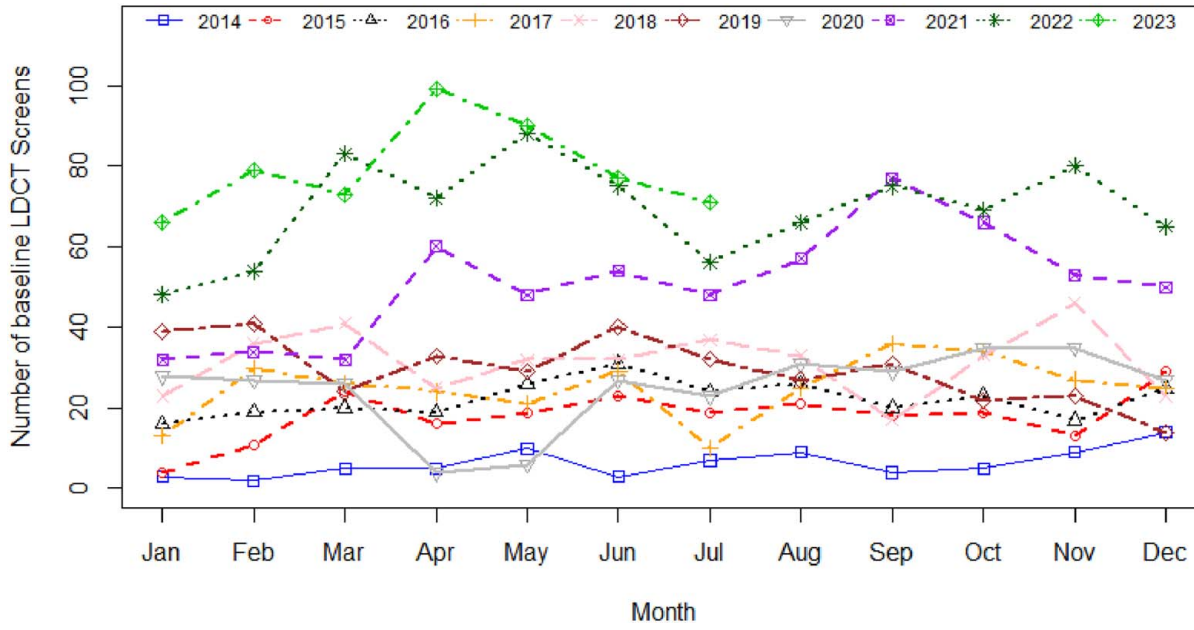


FIGURE 2. The number of new LCS patients by month and year between January 1, 2014, and August 25, 2023. Data from 2013 were excluded because only 3 patients were screened.

Lung Cancer Outcomes Between Subgroups

Lung cancer outcomes are summarized in Table 4. There were 89 and 17 patients with lung cancers in the preguideline and postguideline revision groups, respectively. Eight lung cancers were diagnosed among 943 newly eligible patients in the postguideline revision group. The cancer detection rate was significantly lower in the postguideline revision group compared with the preguideline revision group (0.9% vs. 4.5%, $P < 0.05$). The lung cancer detection rate was higher on the baseline screen (ie, prevalence lung cancers) than on the follow-up screens (ie, combining missed prevalence, incidence, and missed incidence lung cancers) in the pregroup [pre: baseline vs. follow-up 3.5% (68/1963) vs. 1.6% (19/1206), $P < 0.05$]. However, in the postgroup, the higher lung cancer detection rate on the follow-up screens did not reach statistical significance [post: baseline vs. follow-up 0.9% (17/1966) vs. 0% (0/420), $P = 0.05$]. There were no significant differences in the remaining variables. In a sensitivity analysis that included only prevalent lung cancers, the findings remained consistent (Table 5).

DISCUSSION

Our findings shed light on the potential contributing factors behind the surge in LCS participants at our institution. We observed a roughly threefold increase in new patient volume every month immediately following the implementation of the new USPSTF LCS guidelines. Our LCS program's efforts to accommodate the LCS patient influx included opening 3 new imaging locations with 4 additional CT scanners to cater to patients from broader geographical areas and dedicating 2 nurse practitioners for shared decision-making visits. The increase in the proportion of newly eligible patients in the postguideline revision group implies that the surge in new patient volume may be mainly attributed to expanding eligibility criteria. As a result, we observed reductions in mean age, smoking pack-years, PLCOm2012 risk score, and an increase in negative baseline screening findings following the new LCS guidelines. The role of promotional activities, such as media advertising efforts to raise public awareness, in improving awareness and participation in this volume shift remains to be explored.

Racial, ethnic, and sexual disparities persisted under the 2021 USPSTF lung cancer screening guidelines.[22–24] It is possible that the changes in the LCS guidelines address may not address the root causes of the disparities. In our study, simply revising the guidelines increases eligibility but does not guarantee participation in LCS for these populations. In the postguideline revision group, our institution's LCS program reduced the overall proportion of Black patients from 6.9% to 5.1%. This is despite the fact that Black individuals are disproportionately impacted by lung cancer relative to White individuals, facing the highest mortality rates and lowest survival rates for the disease.[25,26] In our cohort, Black patients had the highest average PLCOm2012 risk score in both periods compared with other racial/ethnic groups, which is consistent with prior research.[27] In addition, the proportion of Hispanic patients increased from 5.5% to 7.6%. Nevertheless, considering our institution is in Los Angeles County, where half of the population is Hispanic,[28] ethnic disparities in LCS eligibility and participation remain notable, as demonstrated in previous studies.[13,29] These findings underscore the need for continued emphasis on active outreach and patient education efforts to promote LCS among racial and ethnic minority groups, as well as the further evaluation of how these initiatives impact participation and outcomes.

Similar to previous studies,[30] we observed a higher lung cancer detection rate at baseline than on subsequent screening rounds in the preguideline revision group. Several reasons may explain why the lung cancer detection rate was lower in the postguideline revision group. Compared with the preguideline revision group, participants in the

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TABLE 2. Comparison of Patient Characteristics Pre and Post the new USPSTF Guidelines at Baseline Screen (N = 3929)

Variable	Category	Pre n = 1963	Post n = 1966	*P*
Age [mean (SD)]	-	65 (6)	63 (7)	< 0.001
Sex, n (%)	Female	744 (37.9)	761 (38.7)	0.63
	Male	1219 (62.1)	1205 (61.3)	—
Race/ethnicity, n (%)*	American Indian or Alaska Native	4 (0.2)	14 (0.7)	< 0.001
	Asian	152 (7.7)	166 (8.4)	—
	Black	136 (6.9)	101 (5.1)	—
	Hispanic	108 (5.5)	150 (7.6)	—
	Native Hawaiian or Pacific Islander	1 (0.1)	0 (0)	—
	White	1512 (77.0)	1408 (71.6)	—
	Other	22 (1.1)	13 (0.7)	—
	Unknown	28 (1.4)	114 (5.8)	—
Education level, n (%)	Less than high school graduate	59 (3.0)	69 (2.5)	< 0.001
	High school graduate or GED	283 (14.4)	350 (17.8)	—
	Some training after high school	128 (6.5)	9 (0.5)	—
	Some college or technical school	452 (23.0)	597 (30.4)	—
	College graduate	514 (26.2)	448 (22.8)	—
	Postgraduate	335 (17.1)	331 (16.8)	—
	Unknown	192 (9.8)	162 (8.2)	—
Primary insurance, n (%)†	Medicare and/or Medicaid	843 (42.9)	529 (26.9)	< 0.001
	Private or commercial	1071 (54.6)	1426 (72.5)	—
	Other	49 (2.5)	7 (0.4)	—
	Unknown	0 (0)	4 (0.2)	—
BMI [mean (SD)]	NA	27 (5)	27 (5)	0.22
	Unknown, n (%)	2 (0.1)	1 (< 0.1)	—
Smoking status, n (%)	Current	862 (42.9)	836 (42.5)	0.38
	Former	1098 (55.9)	1128 (57.4)	—
	Unknown	3 (0.2)	2 (0.1)	—
Average number of cigarettes/day [mean (SD)]	NA	22 (9)	19 (7)	< 0.001
	Unknown, n (%)	55 (2.8)	12 (0.6)	—
Smoking duration in years [mean (SD)]	NA	41 (9)	37 (10)	< 0.001
	Unknown, n (%)	53 (2.7)	12 (0.6)	—
Years since quit among former smokers [mean (SD)]	NA	6 (50)	7 (5)	0.06
	Unknown, n (%)	0 (0)	0 (0)	—
Pack-years [mean (SD)]	NA	44 (18)	35 (15)	< 0.001
	Unknown, n (%)	0 (0)	0 (0)	—
Personal history of cancer, n (%)	Yes	313 (15.9)	281 (14.3)	0.16
	No	1650 (84.1)	1685 (85.7)	—
Family history of lung cancer, n (%)	Yes	419 (21.3)	457 (23.2)	0.16
	No	1544 (78.7)	1509 (76.8)	—
COPD, n (%)	Yes	338 (17.2)	265 (13.5)	0.001
	No	1625 (82.8)	1701 (86.5)	—
PLCOm2012 risk score in % [mean (SD)]‡	NA	4.1 (4.0)	2.9 (3.6)	< 0.001
	Unknown, n (%)	276 (14.1)	275 (14.0)	—
Lung-RADS category, n (%)	0	5 (0.3)	8 (0.4)	< 0.001
	1	284 (14.5)	495 (25.2)	—
	2	1374 (70.0)	1263 (64.2)	—
	3	140 (7.1)	100 (5.1)	—
	4A	88 (4.5)	65 (3.3)	—
	4B	52 (2.6)	26 (1.3)	—
	4X	20 (1.0)	9 (0.5)	—
Patient type, n (%)	Prequalified (eligible by 2013 USPSTF guidelines), n (%)	1963 (100)	1023 (52.0)	< 0.001
	Newly eligible (eligible by 2021 but not 2013 USPSTF guidelines), n (%)	0 (0)	943 (48.0)	—

*Subcategories in other race: more than one race, or other racial and ethnic groups not otherwise stated.
†Subcategories in other insurance: Veterans Administration, self-pay, or other insurance not specified.
‡The PLCOm2012 risk score was calculated for patients with complete data in all model covariates.

postguideline revision group had a lower mean PLCOm2012 risk score for lung cancer. An appropriate PLCOm2012 risk threshold for identifying individuals for LCS is a minimum of a 1.5% 6-year risk.[31] Most of our LCS patients in the preguideline revision group met this criterion (25 percentile: 1.6%); however, about half of the patients in the postguideline revision group fulfilled this criterion (median: 1.6%). Moreover, previous studies reported that lung cancer was found in 0.9% to 2.7% at baseline (prevalence) screening and up to 0.6% to 1.5% at annual (incidence) screening.[32] In the postguideline revision group, all identified lung cancers are classified as prevalent lung

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TABLE 3. Mean PLCOm2012 Risk Scores Stratified by Patient Characteristics in the Preguideline and Postguideline Revision Groups

		Pre n = 1687			Post n = 1691		
Variable	**Category**	**n (%)**	**PLCOm2012 mean (SD)***	***P***	**n (%)**	**PLCOm2012 mean (SD)***	***P***
Age	< 65	929 (55)	2.2 (1.5)	< 0.001	969 (57)	1.5 (1.3)	< 0.001
	≥ 65	758 (45)	5.6 (4.6)	—	722 (43)	4.8 (4.6)	—
Sex	Female	638 (38)	4.4 (4.5)	0.13	662 (39)	3.1 (4.0)	0.017
	Male	1049 (62)	3.9 (3.6)	—	1029 (61)	2.7 (3.3)	—
Race/ethnicity†	American Indian or Alaska Native	3 (< 1)	1.9 (0.6)	< 0.001	14 (1)	2.2 (1.8)	< 0.001
	Asian	130 (8)	2.7 (2.5)	—	151 (9)	1.5 (1.5)	—
	Black	113 (7)	5.7 (4.0)	—	91 (5)	4.6 (5.1)	—
	Hispanic	94 (6)	2.1 (2.0)	—	128 (8)	1.1 (1.1)	—
	Native Hawaiian or Pacific Islander	0 (0)	NA	—	0 (0)	NA	—
	White	1347 (80)	4.3 (4.1)	—	1307 (77)	3.1 (3.7)	—
	Other	0 (0)	NA	—	0 (0)	NA	—
Education level	Less than high school graduate	56 (3)	6.0 (6.0)	< 0.001	60 (4)	4.2 (4.2)	< 0.001
	High school graduate or GED	270 (16)	4.9 (4.4)	—	322 (19)	3.5 (4.6)	—
	Some training after high school	121 (7)	5.2 (5.0)	—	9 (1)	7.2 (8.4)	—
	Some college or technical school	432 (26)	4.2 (3.9)	—	569 (34)	3.0 (3.4)	—
	College graduate	488 (29)	3.5 (3.6)	—	420 (25)	2.3 (2.9)	—
	Postgraduate	320 (19)	3.5 (2.8)	—	311 (18)	2.5 (2.9)	—
Primary insurance‡	Medicare and/or Medicaid	709 (42)	5.1 (4.4)	< 0.001	458 (27)	4.6 (4.6)	< 0.001
	Private or commercial	937 (56)	3.6 (3.4)	—	1223 (72)	2.3 (2.9)	—
	Other	41 (2)	3.3 (3.2)	—	6 (< 1)	2.3 (1.9)	—
	Missing	0 (0)	NA	—	4 (< 1)	0.9 (0.4)	—
Smoking status	Current	733 (43)	5.0 (4.6)	< 0.001	723 (43)	3.7 (4.3)	< 0.001
	Former	954 (57)	3.4 (3.2)	—	968 (57)	2.3 (2.8)	—
Family history of lung cancer	Yes	383 (23)	5.9 (5.4)	< 0.001	413 (24)	4.1 (4.6)	< 0.001
	No	1304 (77)	3.6 (3.2)	—	1278 (76)	2.5 (3.1)	—
Lung-RADS category	0	2 (< 1)	3.6 (2.3)	< 0.001	6 (< 1)	4.4 (1.9)	< 0.001
	1	236 (14)	3.4 (3.3)	—	427 (25)	2.1 (2.5)	—
	2	1186 (70)	4.0 (3.6)	—	1085 (64)	3.0 (3.8)	—
	3	125 (7)	4.1 (3.5)	—	90 (5)	4.3 (4.6)	—
	4A	80 (5)	6.6 (6.6)	—	52 (4)	4.8 (4.0)	—
	4B	39 (2)	6.1 (5.1)	—	23 (1)	4.6 (4.0)	—
	4X	19 (1)	7.2 (8.0)	—	8 (< 1)	3.4 (2.9)	—
Patient type	Prequalified (eligible by 2013 USPSTF guidelines)	1687 (100)	4.1 (4.0)	NA	799 (47)	4.2 (4.3)	< 0.001
	Newly eligible (eligible by 2021 but not 2013 USPSTF guidelines)	0	NA	—	892 (53)	1.5 (1.6)	—

* The PLCOm2012 risk score was calculated for patients with complete data in all model covariates. Age was converted to categorical variables for this analysis.

†Subcategories in other race: more than one race, or other racial and ethnic groups not otherwise stated.

‡Subcategories in other insurance: Veterans Administration, self-pay, orand other insurance not specified.

NA indicates not applicable.

cancers. The shorter follow-up time in the postguideline revision group implies that there might be insufficient time to detect the incidence lung cancers. In addition, 8 lung cancers in the postguideline revision group were detected among newly eligible patients, which can be attributed to the implementation of the new LCS guidelines. However, screening lower-risk patients may increase harm, such as overdiagnosis and false positives,[33,34] without significantly reducing lung cancer mortality.

As opposed to using the PLCOm2012 score to select LCS-eligible patients, it was assigned retrospectively to gain insights into the lung cancer risk within our LCS population. Research on the distribution of the PLCOm2012 score among LCS participants is limited.[27] Our data demonstrated a higher average PLCOm2012 score among patients of older age, females, Black patients, less educated, government insurance, current smokers, those with a family history of lung cancer, a higher Lung-RADS score, and prequalified patients. The

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TABLE 4. Comparison of Lung Cancer Outcomes Pre and Post the New USPSTF Guidelines (N = 106)

Variable	Category	Pre n = 89	Post n = 17	*P*
Lung cancer detection rate (% (n/total))	—	4.5 (89/1963)	0.9 (17/1966)	< 0.001
Type of lung cancer, n (%)	Prevalence	68 (76.4)	17 (100)	0.48
	Missed prevalence	9 (10.1)	0 (0)	—
	Incidence	7 (7.9)	0 (0)	—
	Missed incidence	3 (3.4)	0 (0)	—
	Interval	0 (0)	0 (0)	—
	Nonscreening or postscreening	2 (2.2)	0 (0)	—
Histology, n (%)	NSCLC	82 (92.1)	14 (82.4)	0.27
	SCLC	5 (5.6)	3 (17.6)	—
	Metastasis	2 (2.2)	0 (0)	—
TNM stage, n (%)	I-II (early)	64 (71.9)	13 (76.5)	1
	III-IV (late)	21 (23.6)	4 (23.5)	—
	Unknown	4 (4.5)	0 (0)	
Age [mean (SD)]	—	69 (6)	67 (5)	0.32
Sex, n (%)	Female	42 (47.2)	8 (47.1)	1
	Male	47 (52.8)	9 (52.9)	—
Race/ethnicity, n (%)	American Indian or Alaska Native	0 (0)	0 (0)	0.43
	Asian	6 (6.7)	1 (5.9)	—
	Black	9 (10.1)	0 (0)	—
	Hispanic	5 (5.6)	2 (11.8)	—
	Native Hawaiian or Pacific Islander	0 (0)	0 (0)	—
	White	69 (77.5)	14 (82.4)	—
	Other	0 (0)	0 (0)	—
Patient type, n (%)	Prequalified (eligible by 2013 USPSTF guidelines), n (%)	89 (100)	9 (52.9)	< 0.001
	Newly eligible (eligible by 2021 but not 2013 USPSTF guidelines), n (%)	0 (0)	8 (47.1)	—

NSCLC indicates non–small cell lung cancer; SCLC, small cell lung cancer; TNM, tumor, lymph node, and metastasis.

findings were consistent in the preguideline and postguideline revision groups, but need external validation in multicenter cohorts in the future. Moreover, patients with Lung-RADS 1 or 2 results (clinically negative results) had a lower mean PLCOm2012 score than those with Lung-RADS 3 or 4 results (clinically positive results). This trend is consistent with a higher risk of lung cancer for Lung-RADS 1 or 2 than 3 or 4 in Lung-RADS versions 1.0 and 1.1.[35,36] However, within the Lung-RADS 4 category in our cohort, as it increases from 4A to 4B/4X, the mean PLCOm2012 score did not exhibit a

TABLE 5. Comparison of Prevalence Lung Cancer Outcomes Pre and Post the New USPSTF Guidelines (N = 85)

Variable	Category	Pre n = 68	Post n = 17	*P*
Lung cancer detection rate (% (n/total))	—	3.5 (68/1963)	0.9 (17/1966)	< 0.001
Histology, n (%)	NSCLC	62 (91.2)	14 (82.4)	0.48
	SCLC	5 (7.4)	3 (17.6)	—
	Metastasis	1 (1.5)	0 (0)	—
TNM stage, n (%)	I-II (early)	48 (70.6)	13 (76.5)	1
	III-IV (late)	17 (25.0)	4 (23.5)	—
	Unknown	3 (4.4)	0 (0)	—
Age [mean (SD)]	—	68 (6)	67 (5)	0.40
Sex, n (%)	Female	34 (50.0)	8 (47.1)	1
	Male	34 (50.0)	9 (52.9)	—
Race/ethnicity, n (%)	American Indian or Alaska Native	0 (0)	0 (0)	0.40
	Asian	4 (5.9)	1 (5.9)	—
	Black	8 (11.8)	0 (0)	—
	Hispanic	4 (5.9)	2 (11.8)	—
	Native Hawaiian or Pacific Islander	0 (0)	0 (0)	—
	White	52 (76.5)	14 (82.4)	—
	Other	0 (0)	0 (0)	—
Patient type, n (%)	Prequalified (eligible by 2013 USPSTF guidelines), n (%)	68 (100)	9 (52.9)	< 0.001
	Newly eligible (eligible by 2021 but not 2013 USPSTF guidelines), n (%)	0 (0)	8 (47.1)	—

NSCLC indicates non–small cell lung cancer; SCLC, small cell lung cancerTNM, tumor, lymph node, and metastasis.

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monotonically increasing pattern (ie, mean PLCOm2012 score: 4B/X > 4A) in both groups. Besides, the American College of Radiology Lung-RADS Committee removed the Risk of Malignancy column from the Lung-RADS 2022 table, such that risk assessment is now lesion specific, except for category 4B lesions. Given this, more research is needed to verify the correlation between the positive Lung-RADS categories and the PLCOm2012 risk scores in more diverse and larger cohorts.

Multiple confounding factors may influence the study. There was a tremendous increase in the proportion of privately insured patients in the postguideline revision group than the pregroup (73% vs. 55%). As a result, the proportion of patients holding government insurance decreased (27% vs. 43%). In the United States, insurance status was associated with several health disparities, such as access to care, quality of care, and prognosis.[37–39] It remained unknown whether such health disparities contributed to the observed changes in our LCS cohort post the guideline revision. Furthermore, the COVID-19 pandemic mainly affected vulnerable populations with preexisting conditions, including lung diseases. The group of individuals who died from COVID-19 may overlap with those at risk of developing lung cancer. For individuals eligible for LCS but who had not yet participated, COVID-19 could have caused their death before they could enter the screening program and potentially develop detectable lung cancer, resulting in a lower prevalence lung cancer detection rate in the postguideline revision group. Even for those who had completed a baseline LCS scan, COVID-19–related deaths might have occurred before they could receive a lung cancer diagnosis, contributing to the low detection rates observed in follow-up screenings in the postgroup.

Limitations

This study has limitations. First, it was not possible to track patients who had permanently moved but continued lung cancer testing and eventually established a lung cancer diagnosis at outside institutions. Therefore, the lung cancer detection rate may be underestimated for both subgroups. Second, we report on experience from a single health care setting; our findings might not apply to other clinical LCS programs for several reasons, such as differences in patient sociodemographic and smoking variables and staffing and screening capacity constraints to accommodate the increased patient volume. For example, a study by Henderson et al[17] analyzed 17,702 individuals who underwent screening and found that the updated 2021 USPSTF recommendations led to a higher proportion of lung cancer screening participants among non-Hispanic Black individuals and women. This trend, however, was not observed in our lung cancer screening population. Moreover, we were unable to account for relevant factors that may have affected enrollment trends, such as the awareness and acceptance of LCS by health care providers. In addition, the postguideline revision period was shorter than the prephase. A longer follow-up period for the postcohort could provide more meaningful insights into the long-term effects of the guidelines on lung cancer outcomes, which may have been underrepresented due to the short observation period in this study. For instance, despite a similar lung cancer detection rate among the prequalified and newly eligible patients in the postperiod observed in our cohort, we would expect a higher lung cancer detection rate in the prequalified patients than in the newly eligible patients in the long run. Last but not least, further analysis could clarify racial/ethnic disparities by estimating screening rates over time among eligible individuals in each group, complementing the existing data on group proportions among those screened.

CONCLUSIONS

Following adopting the 2021 USPSTF LCS guidelines, our program experienced an approximately 3-fold surge in monthly patient volume. The expanded eligibility criteria led to the detection of lung cancers in newly eligible patients; however, the proportion of screened patients with lung cancer decreased. Despite lower eligible age and smoking intensity, disparities in LCS participation persisted among certain racial and ethnic groups. This indicates that additional efforts in active outreach and patient education are required to promote greater engagement in LCS.

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